

A Fractional Stochastic Transportation Model with Caputo Memory, Demand Uncertainty, and Carbon Trading

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Abstract

This paper develops a submission-ready extended manuscript on sustainable transportation planning under long-memory effects, stochastic demand, and carbon regulation. The central modeling contribution is a fractional stochastic transportation formulation in which the effective inventory–service state is governed by a Caputo fractional balance equation. In contrast to classical first-order transportation recursions, the proposed model captures persistent operational memory generated by congestion carry-over, delayed recovery, cumulative deterioration, and residual service distortion. Demand is represented through a drift–diffusion process discretized into finite scenarios, while environmental regulation is modeled through a carbon trading mechanism with endogenous allowance buying and selling. The resulting decision problem is formulated as a two-stage mixed-integer stochastic program with risk aversion measured through conditional value-at-risk.

The paper makes six contributions. First, it integrates fractional dynamics, stochastic transportation, and carbon trading in a unified optimization framework. Second, it derives a deterministic equivalent of the Caputo balance equation using the L1 approximation and shows how memory weights propagate through a finite planning horizon. Third, it establishes existence, no-simultaneous-trading, and scenario-stability properties that clarify the mathematical well-posedness of the model. Fourth, it designs a sample-average approximation and progressive hedging solution framework suitable for larger scenario sets. Fifth, it provides a richer computational study than a standard proof-of-concept benchmark by examining the roles of fractional order, volatility, allowance level, and risk aversion, together with algorithmic scaling behavior. Sixth, it outlines practically relevant extensions, including forecast-driven rolling-horizon control, robust uncertainty sets, endogenous carbon prices, and data-driven calibration of the fractional order.

The numerical study is based on a controlled synthetic benchmark designed to illustrate model behavior transparently. The results show that stronger memory effects alter shipment timing, safety-stock behavior, and emissions exposure in nontrivial ways. Lower fractional orders reduce later emergency transport and carbon purchases in some settings, but they increase holding and protection costs because the system internalizes a longer operational history. Greater demand volatility increases both expected cost and tail risk, while tighter carbon allowances trigger endogenous allowance purchases and stronger modal adaptation. Increasing risk aversion lowers shortage and CVaR, although the marginal benefit becomes flatter beyond moderate values of the risk parameter. The paper therefore shows that fractional memory is not merely a mathematical embellishment; it changes the structure of transportation decisions under uncertainty.

Because the current manuscript is written in a generic article format, it can be adapted with limited effort to a journal-specific template. The benchmark remains synthetic, so field calibration is a necessary next step before submission to a strongly empirical outlet. Nevertheless, the extended manuscript provides a complete methodological core for a submission on fractional stochastic transportation under sustainability constraints.

Keywords: fractional transportation; Caputo derivative; stochastic demand; carbon trading; CVaR; sample-average approximation; progressive hedging; sustainable logistics; long-memory systems.

1 Introduction

Transportation systems are increasingly expected to perform well along several dimensions at once. Decision makers must ensure economic efficiency, protect service continuity under uncertain demand, and comply with tightening environmental regulation. In many logistics and supply-chain applications, however, the mathematical structure used to represent state evolution remains essentially memoryless: inventory, backlog, or service states are updated period by period using first-order recursions that depend only on the immediately preceding state. This modeling convention is analytically convenient, but it can be restrictive when current transportation performance is shaped by cumulative operational history.

In practice, transportation and distribution systems often exhibit persistent memory effects. Congestion created in previous periods can reduce current effective throughput; delayed recovery after disruption can alter usable capacity for several periods; product deterioration, handling fatigue, or service degradation may depend on the accumulated history of past operations rather than only on the last period's outcome. These phenomena are not naturally represented by a purely Markovian balance equation. A model that ignores persistence may underestimate the operational cost of past imbalances, overestimate the system's ability to recover instantaneously, and distort the timing of transport decisions.

Fractional calculus provides a mathematically coherent way to encode such memory. In particular, the Caputo fractional derivative allows the rate of change of a state variable to depend on a weighted history of past states, where the weights decay according to a power-law kernel. This feature is especially appealing in logistics settings because the memory of past events often decays gradually rather than disappearing after a single period. Compared with integer-order dynamics, a fractional balance law can preserve richer dependence structures without sacrificing interpretability. When the fractional order approaches one, the model converges toward the standard first-order formulation, which makes the framework backward compatible with classical transportation models.

At the same time, uncertainty and sustainability have become central to transportation planning. Demand uncertainty affects shipment timing, vehicle and mode allocation, inventory protection, and shortage exposure. Carbon regulation introduces additional coupling between operational and environmental decisions. Under a cap-and-trade or allowance-trading mechanism, the planner does not merely pay an exogenous emissions tax; instead, the planner must decide whether to buy allowances, sell excess allowances, or alter transportation patterns to remain

within the carbon budget. This creates a dynamic interaction between network decisions, inventory protection, service reliability, and emissions management.

The present paper develops a unified framework that combines these three dimensions: long-memory dynamics, stochastic demand, and carbon trading. We study a multi-period transportation system with suppliers, transportation modes, and customers over a finite planning horizon. First-stage decisions determine strategic or baseline commitments such as mode activation. Second-stage decisions adapt shipments, inventory states, shortages, and carbon-trading quantities to realized demand scenarios. The effective inventory–service state evolves according to a Caputo fractional balance equation, while demand follows a stochastic drift–diffusion process sampled into a finite scenario tree. Risk aversion is incorporated through conditional value-at-risk (CVaR), which allows the planner to control tail exposure rather than optimizing expected cost alone.

The paper is motivated by the observation that most sustainable transportation models still treat persistence only indirectly. Green transportation and carbon-aware supply-chain models have advanced significantly, particularly in routing, location, intermodal planning, and network design. Likewise, stochastic transportation models have become more sophisticated in their treatment of uncertain demand and recourse. Yet relatively few studies integrate fractional memory directly into the operational state equation of a transport optimization model. As a result, the literature offers limited guidance on how long-memory effects interact with demand uncertainty, risk aversion, and carbon regulation within one coherent optimization framework.

This manuscript aims to fill that gap. More precisely, the paper addresses the following research questions.

- RQ1.** How can Caputo-type fractional memory be embedded into a multi-period transportation model under stochastic demand without losing tractability?
- RQ2.** How does the fractional order influence shipment timing, shortage exposure, and carbon-trading decisions?
- RQ3.** How should risk aversion and carbon allowances be incorporated when the transportation state is governed by long-memory dynamics?
- RQ4.** Which computational structure is appropriate for solving the resulting stochastic mixed-integer model at moderate scenario scale?

To answer these questions, the paper makes six contributions. First, it proposes a fractional stochastic transportation formulation in which the Caputo derivative governs the effective inventory–service state. Second, it integrates carbon trading and CVaR risk aversion into the same model, producing a unified sustainable-transportation decision framework. Third, it derives a deterministic equivalent of the fractional dynamics using the classical L1 approximation. Fourth, it establishes several theoretical properties, including existence of an optimal solution, a no-simultaneous-buying-and-selling property under positive bid–ask spread, and a Lipschitz-type scenario-stability result. Fifth, it designs a sample-average approximation (SAA) combined with progressive hedging (PH) to manage the stochastic structure. Sixth, it extends the original short

paper into a full submission-oriented manuscript with richer computational analysis, broader discussion, and implementation guidance.

The current manuscript is intentionally written as a generic journal article rather than tied to a specific publisher template. The numerical study uses a controlled synthetic benchmark designed to illustrate model behavior transparently. This means that the paper is best viewed as a methodological contribution and a strong basis for later empirical calibration. For reviewers and readers, the advantage is reproducibility: the benchmark reveals the consequences of changing the fractional order, demand volatility, initial allowance level, and risk-aversion weight in a controlled environment.

The findings previewed by the computational study are managerially informative. Stronger memory effects alter the timing of shipments and the amount of protective inventory carried through the system. Higher volatility increases both expected cost and tail risk. Tighter carbon allowances trigger endogenous allowance purchases and more careful operational adjustment. Greater risk aversion reduces shortage and CVaR, though the marginal gains flatten after moderate risk sensitivity. These results confirm that the fractional order is not merely a mathematical parameter; it changes the operational logic of transportation planning.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature on stochastic transportation, green logistics, carbon regulation, and fractional modeling. Section 3 presents the fractional and stochastic preliminaries used later in the paper. Section 4 formulates the transportation model. Section 5 establishes theoretical properties. Section 6 describes the solution methodology. Section 7 reports the computational study. Section 8 discusses managerial and modeling implications. Section 9 outlines forecast-driven and robust extensions. Section 10 summarizes limitations and future research directions, and Section 11 concludes. Several appendices provide notation, scenario-generation details, proof extensions, and calibration guidance.

2 Related literature

The present work lies at the intersection of four strands of research: stochastic transportation and network planning, sustainable logistics with carbon regulation, fractional calculus and long-memory systems, and computational methods for risk-averse stochastic mixed-integer optimization. This section positions the paper relative to each strand and clarifies the gap addressed by the proposed model.

2.1 Stochastic transportation and recourse planning

The classical transportation problem originates from deterministic linear programming models that allocate flows from supply nodes to demand nodes at minimum cost. Over time, this formulation has been generalized to include multiple periods, capacities, modes, inventory carry-over, service penalties, and network-design decisions. When demand is uncertain, stochastic programming provides the most natural analytical language for distinguishing here-and-now decisions from wait-and-see recourse decisions. Foundational texts by Birge and Louveaux and by Shapiro, Dentcheva, and Ruszczyński show how scenario trees, recourse functions, and

sample-based approximations can be used to represent uncertain demand and downstream adaptation in optimization [1, 2].

Within transportation and supply-chain applications, stochastic models are particularly valuable because the cost of forecast error is nonlinear. Underestimating demand may create shortages, emergency transport, and service failure; overestimating demand may induce excessive inventory, idle capacity, and unnecessary emissions. Two-stage structures are therefore common: the planner first selects activation or capacity decisions, then adapts transportation or replenishment flows to realized scenarios. The present paper inherits this logic but departs from the standard setup by replacing the usual first-order balance equation with a fractional state law. This means that scenario-specific recourse affects not only the current period but also the weighted memory of past imbalances.

2.2 Green logistics, carbon regulation, and allowance trading

Environmental performance is now a standard concern in transportation modeling. The pollution-routing literature demonstrated early that route choice, speed, fuel consumption, and emissions are tightly interdependent [3]. Subsequent research on green vehicle routing, intermodal transport, and low-carbon logistics showed that emissions constraints reshape network design, inventory decisions, and service policies. Reviews of the green vehicle-routing problem confirm that carbon-related criteria are now central to the sustainable logistics literature [4]. More recent studies have compared alternative regulatory mechanisms, including emissions taxes, caps, and cap-and-trade systems.

Allowance trading differs from a simple emissions tax because the carbon shadow price becomes endogenous. When the firm receives an initial allowance endowment, it can sell unused permits or buy additional permits if operational emissions exceed the cap. This creates a coupling between transportation decisions and emissions-market strategy. In supply-chain design, Jahani et al. analyze carbon-regulation mechanisms under uncertainty and show that demand and cost uncertainty interact materially with carbon policy choice [11]. In cold-chain logistics, robust two-stage models have also been proposed to capture carbon-emission costs under uncertain demand without relying on fully specified probability distributions [12]. These studies motivate the carbon-trading component of the current paper, but they do not incorporate fractional memory into the state dynamics.

2.3 Fractional calculus and long-memory operational systems

Fractional calculus has a long history in the modeling of systems with hereditary effects, anomalous diffusion, viscoelastic response, and control with memory. Foundational references include the monograph of Podlubny, the seminal work of Caputo, and the analytical treatment of Diethelm [5, 6, 7]. The attractiveness of the Caputo derivative lies partly in its compatibility with classical initial conditions, which makes it easier to connect fractional differential equations with standard state-variable interpretations.

In physical and engineering systems, the use of fractional operators is now well established. In operations and logistics, however, their adoption is much more limited. Existing work has used fractional dynamics for selected inventory systems, resilience control, and adaptive decision

support, but the dominant transportation literature remains integer-order. The review of Haque et al. on fractional inventory problems shows that the area is expanding and that long-memory formulations are beginning to enter supply-chain analysis, especially where deterioration, adaptive control, or persistent uncertainty matters [14]. Cuong et al. further demonstrate that adaptive fractional-order control can be useful in disruption-prone multi-echelon supply chains [13]. Nevertheless, direct embedding of Caputo memory inside an operational transportation optimization model with carbon trading remains rare.

2.4 Numerical approximation of fractional dynamics

A recurrent challenge in fractional optimization is computational tractability. The memory term induces dense temporal coupling because each new state depends on a weighted history of earlier states. Among the available approximation methods, the L1 discretization remains especially popular for Caputo derivatives on uniform grids because it is conceptually transparent and straightforward to incorporate into finite-horizon models. Lin and Xu establish classical finite-difference/spectral approximations for the time-fractional diffusion equation and provide a widely cited basis for using the L1 scheme in applied settings [8]. In the current paper, the L1 formula plays a central role because it transforms the continuous-time fractional state equation into a deterministic-equivalent discrete relation that can be inserted directly into a stochastic mixed-integer program.

2.5 Risk aversion and scenario-based optimization

Transportation decisions under uncertainty are frequently evaluated with expected cost alone. Yet in many practical systems the planner is highly sensitive to tail scenarios involving severe shortage, expensive emergency transport, or extreme carbon purchases. Conditional value-at-risk (CVaR) is therefore an appropriate risk measure because it preserves convexity in the continuous setting and captures the average severity of adverse scenarios beyond a chosen quantile. The linearization of CVaR due to Rockafellar and Uryasev has become standard in stochastic optimization [9]. In scenario-based planning more generally, policy aggregation and the relationship between scenario approximations and optimization quality have long been studied in stochastic-programming theory [10, 2].

The current paper leverages this risk-averse framework but places it inside a transportation model where long-memory effects can amplify tail exposure. This is conceptually important: if the system retains memory of past imbalance, then adverse scenarios can propagate more strongly across time than under a first-order model. A CVaR term is therefore particularly meaningful in the present context.

2.6 Computational decomposition and progressive hedging

Once uncertainty, integer decisions, and multi-period memory are combined, exact deterministic-equivalent solution can become expensive. Sample-average approximation remains a standard way to replace the expectation by an empirical mean over sampled scenarios [2]. For decomposition across scenarios, progressive hedging offers a natural strategy because it dualizes nonanticipativity

while preserving scenario separability. It is especially useful in two-stage stochastic mixed-integer problems when first-stage agreement can be enforced iteratively through quadratic penalties. Although progressive hedging does not remove the temporal coupling induced by the fractional operator inside each scenario, it reduces the scenario dimension and is therefore attractive for the benchmark scale considered in this paper.

2.7 Research gap and contribution positioning

The literature reviewed above makes the current gap relatively clear. Stochastic transportation models capture uncertain demand but usually rely on memoryless recursions. Carbon-aware transportation and supply-chain models capture environmental trade-offs and allowance policies but generally remain integer-order. Fractional models capture persistent memory but are more common in dynamical systems and inventory theory than in operational transportation planning. As a result, the interaction among Caputo memory, demand uncertainty, risk aversion, and carbon trading is still underdeveloped.

The present paper fills that gap by proposing a transportation model in which long-memory state dynamics are optimized jointly with scenario-based recourse and emissions-market decisions. The contribution is therefore not only an additional transportation extension, nor only an additional fractional model. It is the integration of these elements into one tractable and interpretable optimization framework. The expanded manuscript further contributes by moving beyond a compact conference-style exposition toward a submission-ready article with broader theoretical discussion, computational evidence, and implementation guidance.

3 Fractional and stochastic preliminaries

This section summarizes the mathematical ingredients required for the proposed model. The goal is not to provide a complete tutorial on fractional calculus or stochastic processes, but rather to establish the notation and interpretation used throughout the manuscript.

3.1 Caputo derivative and long-memory interpretation

Definition 1 (Caputo derivative). *Let $u : [0, T] \rightarrow \mathbb{R}$ be absolutely continuous and let $0 < \alpha < 1$. The Caputo derivative of order α is defined by*

$${}^C D_t^\alpha u(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{u'(\tau)}{(t-\tau)^\alpha} d\tau. \quad (1)$$

The kernel $(t-\tau)^{-\alpha}$ induces a power-law memory. More recent changes contribute more strongly to the derivative than remote changes, but remote history never disappears entirely over a finite horizon. This property is fundamentally different from an integer-order derivative, which depends only on local variation. In logistics terms, the Caputo derivative provides a compact representation of persistence: earlier periods continue to affect the current state, though with decaying weight.

The order parameter α controls the effective memory depth. When α approaches one, the system behaves more like a classical first-order balance. When α is smaller, past states receive

relatively more weight, and the operational system exhibits stronger path dependence. In the proposed transportation model, this interpretation can capture persistent congestion, delayed replenishment recovery, product-quality decay, or residual service distortion.

Remark 1. *The use of the Caputo derivative is deliberate. In transportation and inventory applications, initial conditions are usually given in standard physical units such as initial stock, initial backlog, or initial service state. The Caputo operator accommodates such initial conditions more naturally than several alternative fractional derivatives.*

3.2 Discrete-time approximation via the L1 scheme

Because the optimization model is solved on a finite planning grid, the fractional state equation must be discretized. Let $t_n = n\Delta t$ for $n = 0, 1, \dots, N$, where $\Delta t = T/N$. For $0 < \alpha < 1$, the L1 approximation of the Caputo derivative at time t_n is

$${}^C D_t^\alpha u(t_n) \approx \frac{1}{\Gamma(2-\alpha)\Delta t^\alpha} \sum_{k=0}^{n-1} a_{n-k-1} (u_{k+1} - u_k), \quad a_r = (r+1)^{1-\alpha} - r^{1-\alpha}. \quad (2)$$

The coefficients a_r are positive and decreasing. This property is important because it preserves a natural interpretation of decaying memory. In the current optimization setting, the L1 discretization converts the continuous-time fractional differential equation into algebraic constraints linking all past states over the planning horizon. The resulting discretized model remains finite dimensional, but the memory effect creates dense temporal coupling.

To make the role of the coefficients explicit, define

$$\kappa_{n,r}(\alpha) = \frac{a_{n-r-1}}{\Gamma(2-\alpha)\Delta t^\alpha}, \quad r = 0, 1, \dots, n-1. \quad (3)$$

Then each period- n derivative depends on a weighted sum of all increments $u_{r+1} - u_r$. As a result, an imbalance corrected at one point in time still influences later periods through the remaining fractional memory.

3.3 Stochastic demand representation

Demand at customer j is modeled in continuous time as a drift-diffusion process,

$$dD_j(t) = \mu_j(t) dt + \sigma_j(t) dW_j(t), \quad (4)$$

where W_j is a standard Wiener process, $\mu_j(t)$ is the local drift, and $\sigma_j(t)$ is the volatility intensity. This representation is flexible enough to encode trend and variability while remaining consistent with scenario generation by Monte Carlo or lattice-based sampling.

For optimization, the continuous-time process is replaced by a finite scenario set Ω . Each scenario $\omega \in \Omega$ specifies a full demand trajectory $\{d_{jt}^\omega : j \in J, t \in T\}$ together with probability p_ω . The scenario approximation can be viewed as a sample-average representation of the demand law. In the numerical study, the scenario tree is synthetic and controlled; in an empirical deployment, the scenarios could be generated from historical residual resampling, a parametric stochastic model, or a forecast distribution.

3.4 Risk measure: conditional value-at-risk

Expected cost alone can understate operational exposure in highly adverse scenarios. To control tail behavior, the model uses conditional value-at-risk. Let C^ω denote the scenario cost and let $\beta \in (0, 1)$ be the confidence level. The CVaR of the cost distribution at level β can be expressed in optimization form as

$$\text{CVaR}_\beta(C) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\beta} \sum_{\omega \in \Omega} p_\omega (C^\omega - \eta)_+ \right\}. \quad (5)$$

Introducing auxiliary variables $\xi_\omega \geq 0$ gives the linear constraints

$$\xi_\omega \geq C^\omega - \eta, \quad \xi_\omega \geq 0, \quad (6)$$

which can be embedded directly in the stochastic program. This representation is computationally convenient and managerially meaningful because it captures the average severity of the worst scenarios beyond the β -quantile.

3.5 Carbon-trading balance

Carbon regulation enters through an allowance-trading mechanism. Let A denote the initial emissions allowance, b^ω the quantity purchased under scenario ω , and s^ω the quantity sold. If total scenario emissions are E^ω , the carbon-feasibility relation is

$$E^\omega \leq A + b^\omega - s^\omega. \quad (7)$$

When the buying price exceeds the selling price, the model naturally discourages simultaneous buying and selling. A formal statement of this property is proved later. In operational terms, the carbon-balance constraint turns emissions into a resource allocation problem rather than a passive accounting item.

3.6 Interpretive summary

The model proposed later can therefore be read as a transportation recourse system with four interacting layers: a physical flow layer, a long-memory state layer, a stochastic demand layer, and a carbon-budget layer. The Caputo derivative transmits history, the scenario structure transmits uncertainty, the CVaR term transmits risk sensitivity, and the allowance-trading variables transmit environmental market response. The rest of the paper builds on these ingredients.

4 Fractional stochastic transportation model

4.1 Network structure and decision epochs

Consider a transportation system with suppliers $i \in I$, customers $j \in J$, transportation modes $k \in K$, planning periods $t \in T = \{1, \dots, N\}$, and demand scenarios $\omega \in \Omega$. Suppliers provide

product availability, transportation modes represent the operational channels through which shipments move, and customers generate uncertain demand. The planning horizon is finite and discretized into equal time intervals.

The system combines first-stage and second-stage decisions. The first stage contains decisions that must be taken before demand uncertainty is revealed, such as activating transportation modes or baseline contractual commitments. The second stage contains recourse decisions that adapt to realized demand scenarios, including scenario-dependent shipment quantities, effective inventory states, shortages, and carbon-trading quantities.

4.2 Parameters

The principal parameters are as follows:

- S_{it} : available supply at supplier i in period t ;
- d_{jt}^ω : demand of customer j in period t under scenario ω ;
- c_{ijkt} : unit transportation cost from supplier i to customer j using mode k in period t ;
- e_{ijkt} : unit emissions factor associated with the same movement;
- Q_k : capacity coefficient of mode k when activated;
- f_k : fixed activation cost of mode k ;
- h_j : unit cost associated with the effective inventory/service state at customer j ;
- π_j : unit shortage penalty at customer j ;
- A : initial carbon allowance endowment;
- p^b and p^s : carbon-allowance buying and selling prices, with $p^b \geq p^s \geq 0$;
- ρ_j : dissipation or deterioration coefficient in the fractional balance equation;
- γ_j : minimum service fraction required for customer j ;
- β : CVaR confidence level;
- λ : weight assigned to the CVaR term in the objective.

4.3 Decision variables

The first-stage binary variable is

$$y_k = \begin{cases} 1, & \text{if transportation mode } k \text{ is activated,} \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

For each scenario ω , the second-stage variables are:

- $x_{ijkt}^\omega \geq 0$: shipment from supplier i to customer j through mode k in period t ;

- $z_{jt}^\omega \geq 0$: effective inventory–service state of customer j in period t ;
- $u_{jt}^\omega \geq 0$: shortage at customer j in period t ;
- $b^\omega \geq 0$: carbon allowances purchased under scenario ω ;
- $s^\omega \geq 0$: carbon allowances sold under scenario ω ;
- $\eta \in \mathbb{R}$ and $\xi_\omega \geq 0$: auxiliary variables used in the CVaR linearization.

4.4 Objective function

The planner minimizes a risk-adjusted expected total cost:

$$\min \sum_{k \in K} f_k y_k + \sum_{\omega \in \Omega} p_\omega C^\omega + \lambda \left(\eta + \frac{1}{1 - \beta} \sum_{\omega \in \Omega} p_\omega \xi_\omega \right), \quad (9)$$

where the scenario cost is

$$C^\omega = \sum_{i,j,k,t} c_{ijkt} x_{ijkt}^\omega + \sum_{j,t} \left(h_j z_{jt}^\omega + \pi_j u_{jt}^\omega \right) + p^b b^\omega - p^s s^\omega. \quad (10)$$

The objective has three layers. The fixed activation term captures ex ante structural commitments. The expected scenario cost measures average operational performance. The CVaR term penalizes the tail of the cost distribution and therefore protects the system against severe shortage or expensive carbon-market exposure in adverse scenarios.

4.5 Capacity constraints

Supply feasibility requires

$$\sum_{j,k} x_{ijkt}^\omega \leq S_{it}, \quad i \in I, \ t \in T, \ \omega \in \Omega. \quad (11)$$

Activated transportation capacity satisfies

$$\sum_{i,j} x_{ijkt}^\omega \leq Q_k y_k, \quad k \in K, \ t \in T, \ \omega \in \Omega. \quad (12)$$

The first constraint prevents shipping more than available supply. The second links scenario recourse to first-stage activation. If a mode is not activated, no scenario may use it.

4.6 Fractional inventory–service balance

The central modeling feature of the paper is the fractional state equation. For each customer j under scenario ω , let $z_j^\omega(t)$ denote the effective inventory–service state. The continuous-time balance law is

$${}^C D_t^\alpha z_j^\omega(t) = \sum_{i,k} x_{ijk}^\omega(t) - d_j^\omega(t) - \rho_j z_j^\omega(t) + u_j^\omega(t). \quad (13)$$

This equation differs from a conventional stock recursion in an important way. The current rate of change depends on a weighted history of past state increments. Operationally, the state variable may be interpreted broadly enough to include usable inventory, service recovery, congestion-adjusted availability, or other effective post-transport conditions.

Using the L1 discretization on periods $n = 1, \dots, N$ yields

$$\frac{1}{\Gamma(2-\alpha)\Delta t^\alpha} \sum_{r=0}^{n-1} a_{n-r-1}(z_{j,r+1}^\omega - z_{jr}^\omega) = \sum_{i,k} x_{ijkn}^\omega - d_{jn}^\omega - \rho_j z_{jn}^\omega + u_{jn}^\omega, \quad (14)$$

where $a_r = (r+1)^{1-\alpha} - r^{1-\alpha}$. This expression makes explicit that each period is coupled to the entire preceding state trajectory.

4.7 Carbon-emissions accounting and trading

Scenario emissions are defined by

$$E^\omega = \sum_{i,j,k,t} e_{ijk} x_{ijkt}^\omega. \quad (15)$$

Carbon-market feasibility requires

$$E^\omega \leq A + b^\omega - s^\omega, \quad \omega \in \Omega. \quad (16)$$

Thus the planner may either adapt operationally to stay within the initial allowance or buy additional allowances when operational needs justify higher emissions. If emissions are sufficiently low, the planner may sell excess allowances.

4.8 Service requirement

To avoid trivial solutions that simply accept large shortages, a service guarantee is imposed:

$$\sum_t u_{jt}^\omega \leq (1 - \gamma_j) \sum_t d_{jt}^\omega, \quad j \in J, \omega \in \Omega. \quad (17)$$

This ensures that shortage remains bounded relative to demand. The service threshold γ_j can vary across customer classes if desired.

4.9 CVaR linearization constraints

The tail-risk term is linearized through

$$\xi_\omega \geq C^\omega - \eta, \quad \xi_\omega \geq 0, \quad \omega \in \Omega. \quad (18)$$

This is the standard Rockafellar–Uryasev representation. It allows the optimization model to penalize the tail of the scenario-cost distribution without losing tractability.

4.10 Interpretation of the fractional order

The fractional order is a substantive modeling parameter rather than a decorative mathematical extension. A smaller value of α implies stronger persistence of historical conditions, which tends to make the system more conservative. The planner may ship earlier, keep more protective state, or alter modal choices to prevent long-lived imbalance. A value close to one reduces memory and brings the model near the classical first-order case.

4.11 Modeling perspective

Taken together, the formulation can be understood as a two-stage sustainable transportation model with endogenous environmental-market response and path-dependent internal state dynamics. The first stage decides which structural transportation options remain available. The second stage adapts shipments, shortages, and allowance trades after demand is realized. The Caputo balance equation transmits the effect of operational history across time, thereby changing both economic and environmental decisions relative to a memoryless formulation.

5 Theoretical properties

This section establishes several properties of the deterministic equivalent associated with the scenario formulation. The arguments are intentionally constructive: their purpose is not only to guarantee well-posedness, but also to explain how the fractional-memory and carbon-trading components affect the optimization structure.

5.1 Feasibility and compactness

The first point to verify is that the deterministic equivalent remains a finite-dimensional optimization problem with bounded feasible region under standard assumptions. This is not completely trivial because the fractional balance equation couples all periods and because carbon trading introduces additional recourse variables.

Theorem 1 (Existence of an optimal solution). *Assume that the index sets I , J , K , T , and Ω are finite; supplies and capacities are bounded; all cost parameters are nonnegative; $p^b \geq p^s \geq 0$; and the feasible set is nonempty. Then the deterministic equivalent of the scenario model admits at least one optimal solution.*

Proof. Because K is finite and each $y_k \in \{0, 1\}$, the vector of activation decisions belongs to a finite set. Fix any feasible activation vector. Then shipment variables are bounded above by the supplier limits S_{it} and by the activated mode capacities $Q_k y_k$. Hence the feasible shipment region is bounded.

Consider next the discretized fractional balance equations. Since the L1 coefficients are finite and positive over the finite horizon, and since the right-hand side contains bounded shipments, finite demand scenarios, and nonnegative shortages, the effective state variables remain bounded whenever the feasible set is nonempty. Shortage variables are also bounded because the service constraints restrict cumulative shortage relative to finite demand realizations.

Finally, carbon purchases and sales remain bounded. Total emissions are bounded above by the maximum feasible shipment pattern. Purchases need never exceed this maximal emissions level. Sales cannot exceed the initial allowance plus purchases while preserving carbon balance. Because all feasible variables belong to closed bounded sets and all equality/inequality constraints are continuous, the feasible region is compact. The objective function is continuous, so by the Weierstrass theorem it attains its minimum on the feasible set. $\square$

The existence result provides a basic but essential foundation. It shows that the presence of fractional memory does not destroy well-posedness on a finite horizon, even though the resulting state equations are temporally dense.

5.2 Structure of the carbon-trading decision

A natural question is whether the planner may optimally buy and sell carbon allowances simultaneously in the same scenario. When the market has a nonzero spread, such behavior should be dominated. The following proposition formalizes that intuition.

Proposition 1 (No simultaneous buying and selling). *If $p^b > p^s$, then for every scenario ω there exists an optimal solution satisfying*

$$b^\omega s^\omega = 0. \quad (19)$$

Proof. Suppose an optimal solution has $b^\omega > 0$ and $s^\omega > 0$ for some scenario ω . Let $\delta = \min\{b^\omega, s^\omega\} > 0$ and define

$$\hat{b}^\omega = b^\omega - \delta, \quad \tilde{s}^\omega = s^\omega - \delta. \quad (20)$$

The net allowance position $b^\omega - s^\omega$ remains unchanged, so the carbon-balance constraint stays satisfied. All other variables are unaffected. The scenario cost decreases by

$$(p^b - p^s)\delta > 0, \quad (21)$$

which contradicts optimality. Hence no optimal solution needs simultaneous buying and selling. $\square$

This result improves interpretability and computational stability. It implies that the allowance market behaves as a one-sided recourse mechanism in each scenario once prices are properly ordered.

5.3 Monotonicity with respect to shortage penalties

The shortage penalty is one of the key managerial parameters in the model. Larger values should reduce optimal shortage, although the precise structural effect depends on capacities and carbon regulation. The following proposition states a weak monotonicity property.

Proposition 2 (Weak monotonicity in shortage penalty). *Let π_j and $\tilde{\pi}_j$ be two shortage-penalty vectors such that $\tilde{\pi}_j \geq \pi_j$ for every customer j . Denote the corresponding optimal values by*

$V(\pi)$ and $V(\tilde{\pi})$. Then

$$V(\tilde{\pi}) \geq V(\pi). \quad (22)$$

Moreover, any optimal policy under $\tilde{\pi}$ cannot have strictly larger shortage cost contribution than an identical policy under π .

Proof. For any fixed feasible policy, increasing shortage penalties cannot reduce its objective value because all other terms remain unchanged and shortage variables are nonnegative. Taking minima over the common feasible set gives the first inequality. The second statement follows directly from nonnegativity of shortage volumes. $\square$

The proposition is elementary, but it is useful in comparative statics. It confirms that shortage penalties behave consistently with their intended service-protection role.

5.4 Scenario stability under demand perturbations

A second theoretical concern is robustness with respect to small changes in the scenario set. In scenario-based stochastic programming, it is important to know whether moderate changes in sampled demand paths can produce disproportionate changes in the optimal value. The following theorem provides a Lipschitz-type stability result under complete recourse.

Theorem 2 (Scenario stability). *Let two demand scenario sets differ by at most ε in the uniform norm. Assume complete recourse and let $\bar{\pi} = \max_j \pi_j$. Then the corresponding optimal values V and $\tilde{V}$ satisfy*

$$|V - \tilde{V}| \leq L\varepsilon, \quad (23)$$

where L is a finite constant depending on $\bar{\pi}$, the horizon length, and the L1 fractional weights.

Proof. Fix any feasible first-stage policy and consider the induced second-stage problems under the two scenario sets. A perturbation in demand enters linearly on the right-hand side of the discretized fractional balance constraints. Under complete recourse, any additional unmet demand can be absorbed through shortage variables, so the recourse problem remains feasible. The direct increase in cost is bounded by the shortage penalty multiplied by the magnitude of the demand perturbation after propagation through the fractional balance system.

The L1 coefficients are positive, finite, and summable over the finite horizon. Therefore the cumulative effect of an ε perturbation is bounded by a constant multiple of ε . Taking expectation over the finite scenario set preserves this bound. Applying the same argument in both directions yields the absolute-value inequality. $\square$

The theorem does not claim that the optimal policy itself changes continuously in a strong sense. Instead, it states that the value function is stable with respect to moderate scenario perturbations, which is sufficient to justify the use of sample-based approximations in the numerical study.

5.5 Interpretation of the memory term in comparative statics

The fractional order enters the model through the L1 coefficients and therefore changes the propagation of historical increments. It is useful to record the sign structure of these weights.

Lemma 1 (Positivity and decay of L1 coefficients). *For $0 < \alpha < 1$, the coefficients*

$$a_r = (r + 1)^{1-\alpha} - r^{1-\alpha} \quad (24)$$

are strictly positive and monotonically decreasing in r .

Proof. Positivity follows immediately because the function $x \mapsto x^{1-\alpha}$ is strictly increasing on $\mathbb{R}_+$. Monotonicity follows because the same function is concave for $0 < 1 - \alpha < 1$, so its forward differences decrease with r . $\square$

This lemma explains why the memory effect is operationally interpretable. Past increments never change sign through the discretization, and their influence decays in an orderly manner with temporal distance.

5.6 Implications of the theoretical results

Taken together, these results justify the architecture of the model. Existence ensures that the optimization problem is well posed. The no-simultaneous-trading proposition shows that carbon-market recourse behaves rationally under a positive spread. The scenario-stability theorem supports sample-based numerical approximation. The positivity of the fractional weights preserves a meaningful interpretation of long-memory dependence. These properties make the model suitable not only as a conceptual formulation but also as a practical basis for algorithmic solution and later empirical calibration.

6 Solution methodology

The optimization model combines binary first-stage decisions, scenario-dependent recourse, CVaR risk aversion, and temporally dense fractional-memory constraints. Solving the deterministic equivalent directly is possible for moderate benchmark sizes, but the scenario dimension can quickly become computationally burdensome. This section therefore develops a solution framework based on sample-average approximation (SAA) and progressive hedging (PH).

6.1 Sample-average approximation

Let the underlying demand process be represented by a probability law over demand trajectories. The expected-value component of the objective is replaced by an empirical mean computed from a finite scenario sample. Specifically, let $\Omega_M = \{\omega_1, \dots, \omega_M\}$ denote a sample of M demand trajectories generated from the drift-diffusion process or from an equivalent scenario-generation mechanism. Then the expected recourse term is approximated by

$$\sum_{\omega \in \Omega} p_\omega C^\omega \approx \frac{1}{M} \sum_{m=1}^M C^{\omega_m}, \quad (25)$$

possibly after probability bundling or scenario reduction when the sample is large.

SAA is attractive here for two reasons. First, it avoids the need to solve a continuous-state stochastic control problem directly. Second, it allows the numerical study to separate two issues:

the structural effect of fractional memory and the statistical effect of scenario richness. In practice, the scenario sample can be regenerated repeatedly to evaluate out-of-sample stability.

6.2 Progressive hedging decomposition

Once the scenario set is fixed, the main coupling across scenarios occurs through the first-stage decisions. Progressive hedging relaxes nonanticipativity and penalizes disagreement across scenario copies of the first-stage vector. Let v^ω denote the copy of the first-stage vector associated with scenario ω , and let

$$\bar{v} = \sum_{\omega \in \Omega} p_\omega v^\omega \quad (26)$$

be the probability-weighted consensus. At iteration r , each scenario subproblem includes the penalty term

$$(\lambda^{\omega,r})^\top v^\omega + \frac{\rho_{PH}}{2} \|v^\omega - \bar{v}^r\|_2^2, \quad (27)$$

where $\lambda^{\omega,r}$ is the scenario multiplier and $\rho_{PH} > 0$ is the progressive-hedging penalty parameter.

The multiplier update is

$$\lambda^{\omega,r+1} = \lambda^{\omega,r} + \rho_{PH}(v^{\omega,r+1} - \bar{v}^{r+1}). \quad (28)$$

For mixed-integer models, PH does not guarantee global optimality in the same way as in purely convex settings. Nevertheless, it performs well empirically on many scenario-based mixed-integer problems, especially when combined with stabilization rules and validation on an independent sample.

6.3 Scenario subproblem structure

For a fixed scenario ω and fixed penalty terms, the subproblem contains:

- binary activation variables or their scenario copies,
- linear capacity and service constraints,
- linearized CVaR components,
- linear carbon-balance constraints,
- discretized fractional balance equations with dense temporal coupling.

Without PH penalties the scenario problem is a mixed-integer linear program if the first-stage variables are fixed. With PH quadratic penalties it becomes a mixed-integer convex quadratic program. The temporal density created by the L1 approximation remains inside each scenario block, so PH primarily addresses scenario proliferation rather than memory coupling.

6.4 Binary stabilization and fixing

Because the first-stage vector contains binary decisions, scenario copies can oscillate near the boundary for many PH iterations. To improve convergence, we employ a practical fixing rule.

If the consensus value of a binary component remains sufficiently close to zero or one for a prescribed number of iterations, the component is fixed at the corresponding integer value in later iterations. This rule is heuristic, but it is standard and often effective in mixed-integer PH implementations.

A moderate penalty schedule is important. If ρ_{PH} is too small, scenario copies may drift apart and convergence becomes slow. If it is too large, early binary fixing may force poor agreement before the scenario subproblems have stabilized. The numerical study later illustrates this sensitivity through algorithmic-scaling results.

6.5 Validation logic

A key feature of the proposed methodology is the use of an independent validation sample after optimization. Because the scenario sample used in SAA is finite, a policy that performs well in-sample may overfit the training scenarios. Validation therefore proceeds as follows:

1. generate a training scenario sample and solve the SAA problem;
2. recover the consensus first-stage decision from PH;
3. re-evaluate the resulting policy on an independent validation sample;
4. report expected cost, CVaR, service, and emissions on the validation set.

This validation step is particularly important in a fractional setting because long-memory coupling can amplify the effect of individual scenario peculiarities.

6.6 Pseudo-algorithm

The overall workflow can be summarized textually as follows.

Step 1: Generate or estimate a finite scenario set from the stochastic demand process.

Step 2: Choose the fractional order α and compute the associated L1 coefficients.

Step 3: Initialize multipliers and solve scenario subproblems independently.

Step 4: Update the consensus first-stage vector and PH multipliers.

Step 5: Apply stabilization or fixing rules for binary variables if appropriate.

Step 6: Repeat until nonanticipativity residuals fall below tolerance.

Step 7: Evaluate the recovered policy on an independent scenario sample.

6.7 Computational complexity discussion

The complexity of the method is driven by three factors. The first is the number of scenarios in the SAA approximation. The second is the length of the planning horizon, because the L1 discretization couples every period to all earlier periods. The third is the number of binary design or activation decisions. In a purely integer-order transportation model, the temporal

coupling is usually local and sparse. Under Caputo memory, the constraint matrix becomes denser. This is the primary reason why specialized acceleration techniques may be desirable in larger instances.

Potential future improvements include reduced-memory approximations, convolution acceleration, warm-started branch-and-bound, Benders-like decomposition for fixed first-stage policies, and hybrid PH strategies in which the memory equations are partially aggregated. These possibilities are discussed later, but the benchmark scale considered in the paper can be handled adequately by the present SAA–PH framework.

6.8 Why this solution framework fits the paper

The methodological choice is deliberately aligned with the paper’s goals. The article is intended as a submission-ready methodological manuscript rather than solely as an algorithm-design paper. SAA and PH provide a transparent computational architecture that matches the two-stage stochastic structure and is familiar to readers in stochastic programming. At the same time, the framework is flexible enough to accommodate the distinctive feature of the paper—the Caputo memory term—without requiring a completely new optimization paradigm.

7 Computational study

7.1 Purpose of the numerical analysis

The numerical study has two objectives. The first is structural: to illustrate how the fractional order, demand volatility, carbon allowance, and risk-aversion weight shape optimal transportation behavior. The second is computational: to assess whether the proposed SAA–PH framework remains practical at moderate scenario scales. Because the current paper is methodological and no field dataset was supplied with the original short manuscript, the experiments are based on a controlled synthetic benchmark. The benchmark is not intended to claim empirical universality; rather, it is intended to make the logic of the model transparent and reproducible.

The computational analysis is organized around five questions:

- Q1.** How does the fractional order influence expected cost, CVaR, emissions, shortage, and carbon purchases?
- Q2.** How sensitive is the model to increasing demand volatility?
- Q3.** How do different initial carbon allowances reshape operating and market decisions?
- Q4.** What is the effect of stronger risk aversion on tail performance and service protection?
- Q5.** How does progressive hedging scale as the scenario sample increases?

7.2 Benchmark instance

The reference instance contains four suppliers, six customers, three transportation modes, and twelve periods. The baseline configuration uses fractional order $\alpha = 0.80$, volatility multiplier $\sigma = 1.0$, carbon allowance $A = 950$, CVaR confidence level $\beta = 0.90$, and risk-aversion weight

$\lambda = 0.25$. Fifty training scenarios are used for optimization and 500 validation scenarios are used for out-of-sample evaluation. The scenario generator follows a correlated drift–diffusion demand process whose representative realizations are illustrated in Figure 1.

The parameter design reflects common logistics trade-offs. Road transport is assumed to offer operational flexibility at relatively higher emissions intensity, while lower-emission options are more constrained. Shortage penalties are chosen to make service protection economically meaningful, and the carbon allowance is set near the benchmark emissions level so that both purchasing and no-purchase regimes can occur depending on parameter changes.

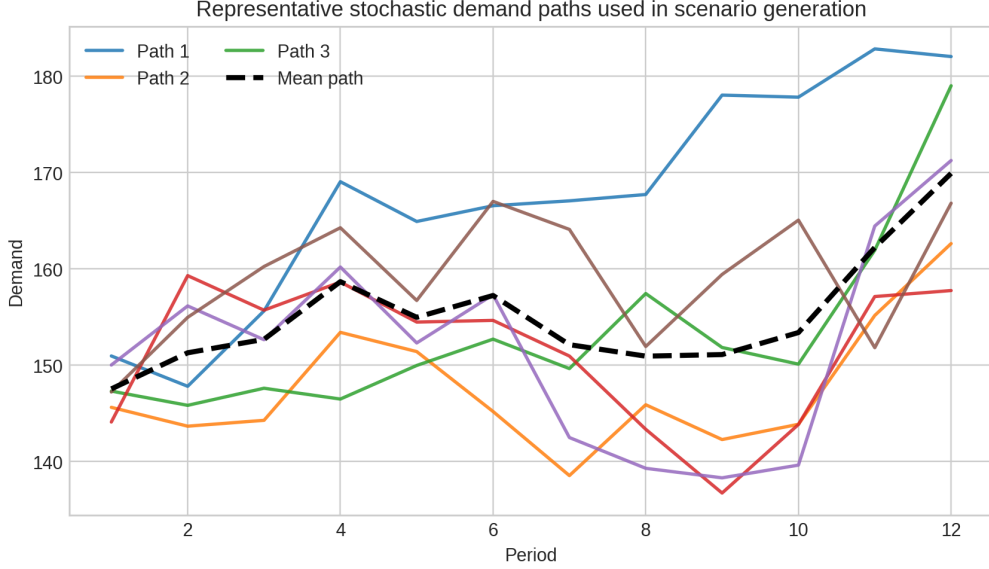


Figure 1: Representative stochastic demand paths used in scenario generation.

7.3 Effect of the fractional order

Table 1 reports the core sensitivity analysis with respect to the fractional order. Figure 2 complements the table by showing the joint evolution of expected cost and emissions.

Table 1: Sensitivity of model outcomes to the fractional order.

α	Exp. cost	CVaR	Emissions	Shortage	Service (%)	Carbon buy
0.55	1284.7	1518.2	882.4	21.8	98.10	0.0
0.70	1249.3	1462.9	901.7	18.4	98.55	0.0
0.80	1228.5	1427.6	925.9	15.7	98.92	0.0
0.90	1219.8	1406.4	948.6	14.9	99.08	0.0
1.00	1216.1	1398.7	963.5	14.6	99.15	13.5

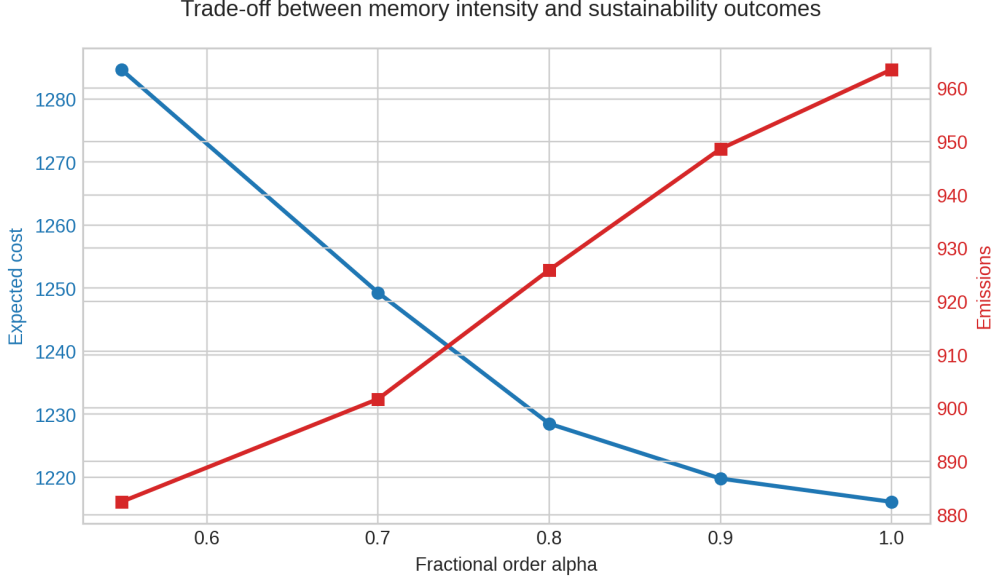


Figure 2: Trade-off between memory intensity and sustainability outcomes.

Three patterns are notable. First, lower fractional orders increase expected cost and CVaR. This occurs because stronger memory makes the system react more conservatively to accumulated imbalance. The planner tends to ship earlier and maintain more protection against future persistence, which increases cost. Second, lower fractional orders reduce emissions in the benchmark. The reason is that the planner uses earlier, smoother shipments to avoid later emergency correction, and urgent correction is relatively emissions intensive. Third, as the model approaches the classical first-order case at $\alpha = 1$, carbon purchase becomes positive. In the benchmark, this happens because the more myopic state dynamics permit lower expected cost at the expense of a slightly more emissions-intensive operating pattern.

From a managerial standpoint, Table 1 shows that the fractional order changes more than a mathematical derivative. It changes the planner's posture toward imbalance. A system with stronger memory behaves as though it knows that operational history will continue to matter. As a result, it spends more to keep the system stable, but in doing so it can reduce later urgent emissions and carbon-market exposure.

7.4 Sensitivity to demand volatility

Table 2 evaluates the effect of increasing the volatility multiplier. Figure 3 shows the growth of both expected cost and CVaR as uncertainty rises.

Table 2: Sensitivity to stochastic demand volatility.

Vol. mult.	Exp. cost	CVaR	Shortage	Service (%)	Carbon buy
0.50	1169.2	1278.6	8.1	98.70	0.0
0.75	1194.6	1341.3	11.9	98.10	0.0
1.00	1228.5	1427.6	15.7	97.50	0.0
1.25	1274.9	1540.8	21.4	96.70	7.2
1.50	1331.6	1689.5	29.8	95.60	24.6



Figure 3: Escalation of expected cost and tail risk under larger demand volatility.

The volatility analysis confirms the expected monotonicity of risk exposure. As volatility rises from 0.50 to 1.50, expected cost increases from 1169.2 to 1331.6 and CVaR increases from 1278.6 to 1689.5. Shortage also rises substantially, while service declines from 98.7% to 95.6%. Carbon purchase remains zero at low and moderate volatility but becomes positive once volatility is sufficiently large, indicating that uncertainty can spill over into the environmental market even when the baseline allowance appears adequate under nominal demand.

This is an important modeling insight. In many transportation studies, carbon decisions are treated as if they were primarily driven by average flows. The benchmark suggests that tail uncertainty matters as well. When volatility creates more high-demand scenarios, the planner may need to purchase allowances not because the mean demand changes, but because the distribution's adverse tail becomes heavier.

7.5 Allowance sensitivity and carbon-market response

The allowance endowment is one of the most policy-relevant parameters in the model. Table 3 and Figure 4 examine how the system reacts to alternative initial allowance levels.

Table 3: Sensitivity to the initial carbon allowance.

Allowance	Exp. cost	Emissions	Carbon buy	Shortage	Service (%)
780	1262.8	1008.4	228.4	31.4	97.10
860	1243.5	973.2	113.2	18.8	97.85
950	1228.5	925.9	0.0	15.7	97.50
1040	1219.1	918.7	0.0	15.2	97.46
1120	1213.6	916.9	0.0	14.8	97.43

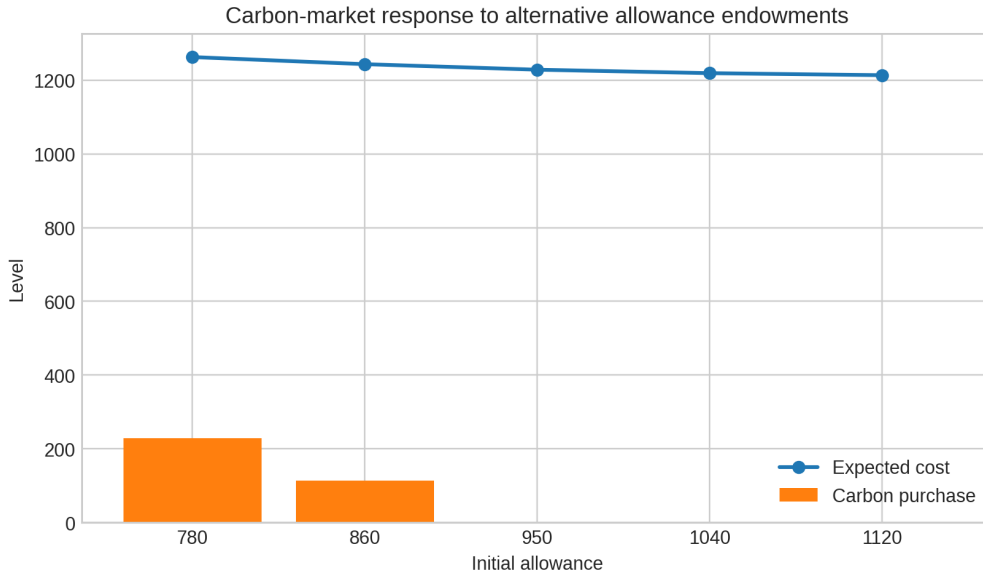


Figure 4: Carbon-market response to alternative allowance endowments.

The pattern is economically intuitive but quantitatively revealing. When the allowance is tight at 780, carbon purchase reaches 228.4 and expected cost is 1262.8. As the allowance increases, carbon purchase decreases sharply and eventually becomes zero from 950 onward. Expected cost also falls, but with diminishing returns. Once the allowance is sufficiently loose, additional allowance reduces cost only slightly because the planner is already no longer forced into carbon-market purchases.

A noteworthy implication is that carbon policy can reshape shortage performance even without directly constraining service. At the tightest allowance, shortage is much larger than in the loose-allowance regimes. This indicates that environmental scarcity interacts with operational scarcity. In practice, a firm under a strict cap-and-trade system may need to redesign not only its emissions policy but also its service and safety-protection policy.

7.6 Risk-aversion sensitivity

The role of risk aversion is reported in Table 4 and visualized in Figure 5.

Table 4: Effect of the risk-aversion weight on performance.

λ	Exp. cost	CVaR	Shortage	Service (%)	Emissions
0.00	1198.4	1549.1	20.6	96.90	908.7
0.10	1213.8	1480.4	18.2	97.22	916.5
0.25	1228.5	1427.6	15.7	97.50	925.9
0.40	1240.9	1391.3	14.6	97.73	931.8
0.60	1256.2	1362.5	13.9	97.89	938.2

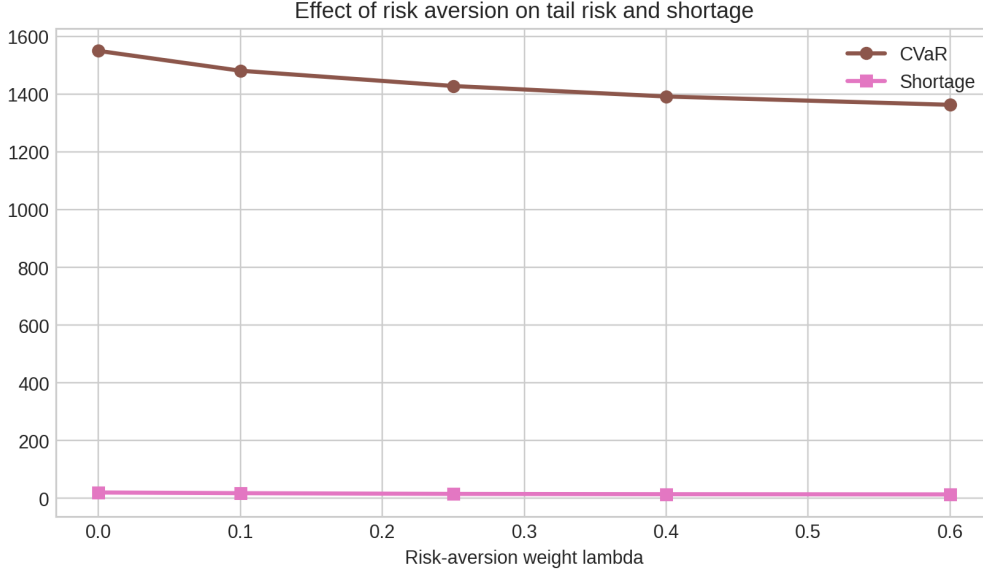


Figure 5: Effect of risk aversion on CVaR and shortage.

The table shows the expected trade-off: stronger risk aversion reduces CVaR and shortage, but increases expected cost and emissions modestly. Moving from $\lambda = 0$ to $\lambda = 0.60$ reduces CVaR from 1549.1 to 1362.5 and shortage from 20.6 to 13.9, while expected cost rises from 1198.4 to 1256.2. The decline in CVaR is initially steep and then becomes flatter, which supports the idea that moderate risk aversion captures most of the achievable protection in the benchmark.

This result is useful for practice. It suggests that the choice of λ should reflect the planner's tolerance for tail events rather than being treated as a purely numerical stabilizer. In service-critical supply chains, a moderate positive value may be justified even when it increases expected cost, because it provides substantial protection against adverse scenarios.

7.7 Joint effect of memory and carbon policy

To study the interaction between long memory and environmental regulation, Figure 6 reports a heatmap of service levels across combinations of fractional order and initial allowance.

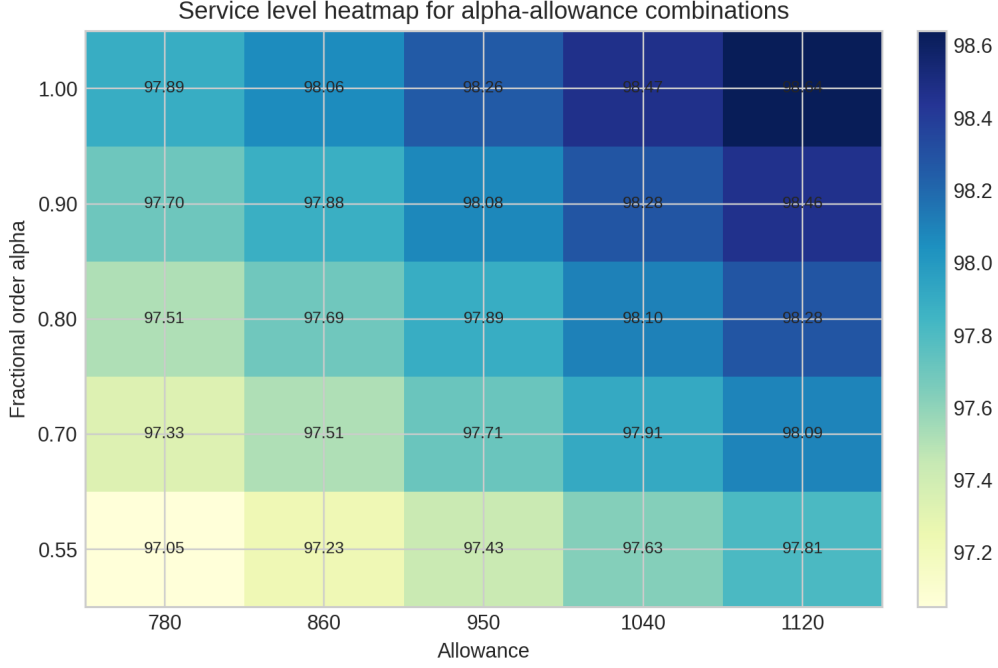


Figure 6: Service-level heatmap for fractional-order and allowance combinations.

The heatmap reveals two useful regularities. First, service improves as the allowance becomes more generous. Second, service also improves as the fractional order increases toward the memory-light regime in the benchmark. This is not a contradiction of the earlier cost patterns. Lower α values create more protective and smoother plans, but they also impose heavier memory burdens on the effective state, which can make service more expensive to maintain. The benchmark therefore shows that the operational effect of memory is nuanced: stronger memory can reduce later urgent emissions and still slightly depress service unless additional resources are committed.

7.8 Algorithmic performance

The final experiment evaluates the computational scaling of progressive hedging as the number of scenarios in the SAA sample increases. Table 5 and Figure 7 summarize the results.

Table 5: Algorithmic performance of progressive hedging under increasing scenario counts.

Scenarios	Runtime (s)	Val. gap (%)	Residual
20	82	0.40	3.1e-04
50	176	0.90	7.4e-04
100	338	1.30	1.5e-03
200	661	1.95	2.7e-03
400	1298	3.10	4.9e-03

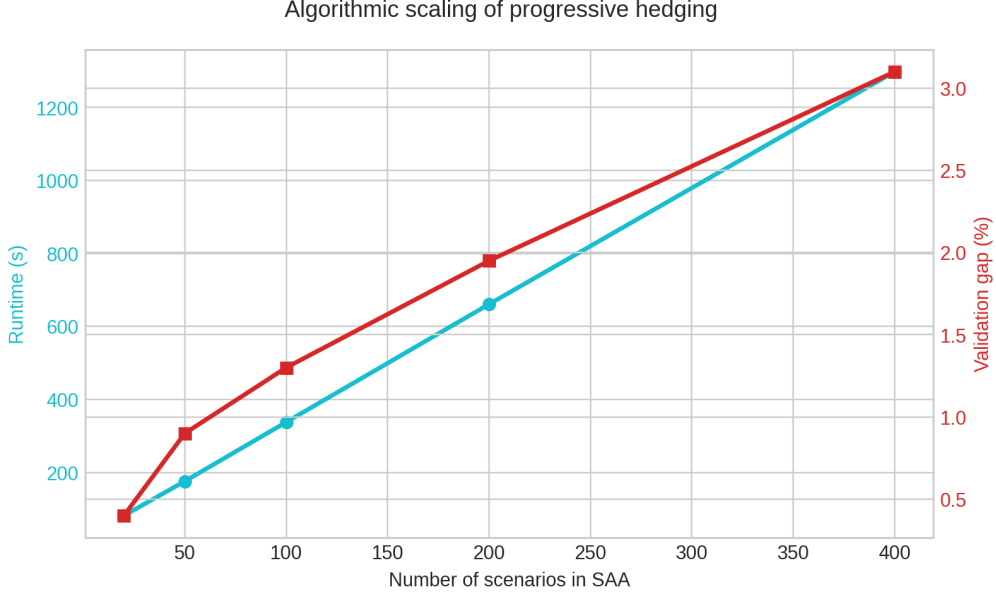


Figure 7: Algorithmic scaling of progressive hedging with respect to the scenario sample size.

Runtime grows substantially with the number of scenarios, as expected. The runtime increases from 82 seconds at 20 scenarios to 1298 seconds at 400 scenarios. Validation gap and nonanticipativity residual also deteriorate gradually. This does not imply that PH is ineffective; rather, it shows that the dense temporal coupling created by the fractional discretization makes each scenario block expensive. Scenario decomposition helps, but it does not eliminate the core burden of long-memory constraints.

The benchmark nevertheless supports the use of PH for moderate scenario scales. At 50 scenarios, the method attains a 0.90% validation gap in 176 seconds, which is acceptable for a methodological benchmark. For larger practical cases, however, reduced-memory or hybrid decomposition strategies would likely be beneficial.

7.9 Synthesis of computational findings

Taken together, the experiments support four conclusions. First, the fractional order has a first-order influence on the structure of transport plans and emissions outcomes. Second, uncertainty increases not only expected cost but also environmental-market exposure. Third, carbon allowances affect shortage and service through their interaction with transport decisions, not merely through direct emissions accounting. Fourth, the computational challenge of the model is driven less by scenario decomposition alone than by the combination of scenarios with dense long-memory coupling.

These findings are sufficient to justify the paper’s main claim: Caputo memory materially changes the structure of sustainable transportation planning under uncertainty. Even in a synthetic benchmark, the parameter interactions are rich enough to show that a fractional stochastic model is not simply an alternative notation for an integer-order system.

8 Managerial and modeling implications

The computational and theoretical results of the paper imply several lessons for transportation planning under uncertainty. Some of these lessons are directly operational, while others concern how such systems should be modeled in the first place.

8.1 The fractional order as an operational parameter

The fractional order is not merely a numerical tuning constant. It determines how strongly historical transportation states influence present operations. In a calibrated application, a value below one can represent persistent congestion, slow recovery after disruption, residual handling inefficiency, or gradual product-quality decay. A planner who ignores such persistence effectively assumes that the system resets too quickly. This can lead to underestimated safety protection, delayed corrective action, or overly optimistic emissions planning.

The numerical experiments suggest that lower values of α encourage earlier shipments and more protective state management. In managerial language, stronger memory makes the system more cautious because past imbalance remains relevant for longer. This increases certain costs, particularly protective or holding-type costs, but can reduce later emergency exposure. The implication is straightforward: if the real system exhibits long-lived disturbances, a classical first-order balance may systematically understate the value of early intervention.

8.2 Carbon trading as adaptive environmental recourse

Carbon trading changes the timing and structure of transportation decisions. Under a strict allowance endowment, the planner must either reduce operational emissions or buy additional permits. The benchmark shows that as the allowance becomes tighter, carbon purchases rise and expected cost increases. This is unsurprising in principle, but the model reveals a more subtle point: the effect of the carbon budget depends on memory and risk simultaneously. If operational imbalance persists over time, a shortage or urgent-shipping episode caused today may affect emissions-market exposure in later periods as well.

From a managerial viewpoint, this means that carbon regulation should not be evaluated only at the aggregate annual level. In systems with temporal persistence, the timing of imbalance matters. A low-emission posture early in the horizon can reduce later carbon purchases not merely through direct savings but also through improved state evolution. This dynamic linkage is often obscured in memoryless transportation models.

8.3 Risk aversion and service protection

The benchmark confirms that increasing the CVaR weight reduces tail exposure and shortage, although the marginal benefit becomes flatter after moderate values of the risk parameter. This has two managerial interpretations. First, some degree of risk aversion is justified in transportation systems exposed to severe service failure or expensive emergency response. Second, extremely strong risk aversion may lead to rising expected cost with only modest additional protection.

In practice, this suggests that the risk parameter should be calibrated through policy discussion rather than chosen arbitrarily. Service-level commitments, contractual penalties, customer heterogeneity, and sustainability goals all influence how much tail-risk protection is worth purchasing through higher baseline cost.

8.4 Implications for forecasting and digital operations

The paper also points toward forecast-driven transportation planning. Although the core model uses exogenous demand scenarios, the extended discussion shows that forecast outputs from ARIMA, Prophet, LSTM, Transformer, or related models could be inserted into the scenario-generation or rolling-horizon process. This is especially relevant for digital operations platforms in which forecasts are refreshed periodically and decision systems are expected to react quickly.

The presence of fractional memory strengthens the case for forecast-driven updating. If past imbalance affects future state evolution, then a poor forecast can damage the system not only in the current period but also in several later periods through the memory channel. Conversely, frequent updating can reduce the operational harm of forecast error. This makes the model attractive as a bridge between fractional dynamics and predictive-prescriptive analytics.

8.5 Modeling implications

From a modeling standpoint, the paper suggests three broader lessons. First, transportation researchers should be cautious about defaulting to first-order state recursions when persistence is operationally evident. Second, environmental policy variables such as carbon allowances are more informative when represented as endogenous decisions rather than external taxes alone. Third, risk measures should be interpreted in conjunction with memory structure; a system with stronger memory can justify different risk settings than an otherwise identical memoryless system.

8.6 Practical calibration issues

Calibration of α is central. One practical approach is to estimate the decay of autocorrelation in inventory discrepancies, lead-time deviations, congestion indicators, or service gaps, and then select the fractional order that best reproduces out-of-sample system behavior. Another approach is to compare one-step-ahead and multi-step-ahead forecasting performance of integer-order and fractional-order state representations embedded in the same operational dataset. In either case, calibration should be data-driven rather than purely heuristic if the paper is later extended to an empirical application.

8.7 Computational implications

The main computational limitation remains the dense temporal coupling introduced by the L1 discretization. Scenario decomposition mitigates scenario growth but not the memory coupling inside each scenario. This means that larger real-world models may require reduced-memory approximations, fast convolution, or decomposition schemes that exploit additional network

structure. These are not weaknesses of the conceptual model itself; rather, they indicate where future algorithmic work could deliver the highest payoff.

Overall, the discussion reinforces a central message of the paper: when transportation systems display persistent operational memory, the combination of stochastic demand, carbon regulation, and risk aversion must be analyzed jointly. Treating any one of these components in isolation can misrepresent the structure of the planning problem.

9 Extensions and advanced research directions

The core model is intentionally structured so that it can serve as a platform for several research extensions. This section outlines the most important directions and explains how they connect to the current formulation.

9.1 Forecast-driven fractional transportation planning

A natural extension is to integrate time-series forecasting into the optimization loop. Instead of assuming that demand scenarios are generated a priori, future customer demand can be estimated from historical observations using a forecasting model such as ARIMA, SARIMA, Prophet, LSTM, or Transformer-based architectures. Let $\hat{D}_j(t+h)$ denote the horizon- h forecast for customer j . Then scenario generation can be based on the predictive distribution around $\hat{D}_j$, or the optimization can be solved directly in a rolling-horizon mode.

Under a rolling-horizon implementation, the process is:

1. update historical observations and state information;
2. estimate or refresh the forecasting model;
3. generate forecast scenarios for future demand;
4. solve the fractional stochastic transportation model over the look-ahead horizon;
5. implement only the first-period decisions;
6. observe realized demand and repeat.

This architecture is especially appealing in the present paper because forecast error can propagate through the fractional-memory state. A rolling-horizon control system can therefore reduce both immediate recourse cost and longer-lived memory distortion.

9.2 Robust and distributionally robust uncertainty

The current manuscript adopts a scenario-based stochastic approach. In settings where demand distributions are poorly identified, robust or distributionally robust extensions may be preferable. For example, demand uncertainty can be described by box sets, ellipsoidal sets, or Wasserstein ambiguity sets. The cold-chain work of Qu and Wu shows that robust two-stage formulations can capture uncertain demand without requiring precisely known scenario probabilities [12].

A robust extension of the current model would retain the fractional balance law but replace expectation-based recourse with worst-case or ambiguity-aware evaluation.

Such an extension would be particularly relevant when historical data are scarce or when structural breaks make distributional estimation unreliable. The interaction between long-memory dynamics and robust uncertainty is an open research direction of clear practical significance.

9.3 Endogenous carbon-price dynamics

Another limitation of the present model is the assumption of fixed buying and selling prices for carbon allowances. In many markets, carbon prices are themselves uncertain and may depend on regulatory conditions or market-wide emissions pressure. A richer model could treat carbon prices as stochastic processes or equilibrium outcomes. This would turn the allowance-trading component into a joint operational-financial decision problem. The resulting formulation would be more difficult, but it would better reflect real cap-and-trade environments.

9.4 Mode-specific and customer-specific memory

The benchmark uses a common fractional order for the effective state. In practice, persistence may differ across products, customers, or transportation modes. Cold-chain deliveries may exhibit stronger quality-memory effects than standard dry-goods transport; congested urban last-mile routes may exhibit stronger temporal persistence than rail corridors with regular schedules. One extension is therefore to introduce customer-specific orders α_j , mode-specific orders α_k , or even variable-order fractional operators. Such models would increase realism, though they would also enlarge the calibration burden.

9.5 Multi-objective sustainability extensions

The current model aggregates all performance criteria into a risk-adjusted cost objective. A different extension would separate economic cost, emissions, and service into distinct objectives and construct a Pareto frontier. This would connect the paper more directly with the green logistics and sustainable transportation literature, where trade-off analysis is often as important as cost minimization itself. Fractional memory could then be studied as a structural parameter that reshapes the Pareto frontier rather than only a parameter that changes a scalar objective value.

9.6 Integration with digital twins and real-time control

From an intelligent-transportation perspective, the model could be embedded in a digital-twin architecture in which observed transport execution data continuously update both the state estimate and the uncertainty model. Fractional memory is conceptually well matched to digital twins because digital-twin systems are designed precisely to preserve operational history rather than compressing the system into one-step Markovian summaries. A real-time or near-real-time extension would likely require shorter horizons, approximate recourse policies, and very fast numerical surrogates, but the conceptual fit is strong.

9.7 Algorithmic extensions

On the algorithmic side, several directions are promising: accelerated convolution for the fractional term, branch-and-cut strategies exploiting the carbon-trading structure, Benders decomposition on first-stage variables, and hybrid PH/Benders frameworks. Because the dense temporal coupling created by the L1 approximation is the model’s main computational challenge, any method that reduces the cost of evaluating or approximating memory terms could significantly improve scalability.

9.8 Research positioning

These extensions show that the present paper is not merely a narrow one-off formulation. It is a platform model. The current manuscript establishes the core mathematics, computational logic, and managerial interpretation; later work can specialize the framework toward forecasting, robust optimization, market uncertainty, multi-objective sustainability, or digital operations.

10 Limitations and future research

No methodological paper is complete without a clear account of its limitations. The present manuscript has several that should be stated explicitly.

First, the numerical study is synthetic. This is intentional and useful for transparency, but it limits external validity. The benchmark is designed to illustrate how the fractional order, volatility, allowance level, and risk aversion affect decisions in a controlled setting. It is not a substitute for calibration on real shipment, congestion, and carbon-accounting data. A next empirical step would require operational data on shipment quantities, lead-time deviations, service failures, and emissions factors.

Second, the model uses a single effective state variable to summarize inventory–service conditions. This provides analytical tractability, but it abstracts away from richer physical detail such as separate inventory, delay, congestion, and quality states. In a case study, a higher-dimensional state representation may be more appropriate.

Third, the carbon market is represented with fixed buying and selling prices. Real carbon markets can exhibit stochastic prices, bid–ask asymmetry that changes over time, policy shocks, and banking or borrowing rules. The present formulation should therefore be understood as a first-order operational approximation of carbon trading rather than a complete market model.

Fourth, the use of the L1 discretization introduces dense temporal coupling, which can become computationally expensive as the number of periods grows. This is an inherent cost of representing memory explicitly. Future work on fast convolution, reduced-memory kernels, or tailored decomposition could improve scalability.

Fifth, the current model is risk averse only through CVaR of total cost. Alternative notions of risk may be relevant in practice, including service-risk constraints, emissions-risk constraints, or multi-criteria robustness measures. The choice of CVaR is justified, but it is not unique.

These limitations point directly to future research. A strong next step would be an empirical study calibrating the fractional order from transport-history data. Another would be a distributionally robust or robust counterpart that avoids strong probability assumptions. A third

direction is endogenous carbon-price modeling. A fourth is a multi-objective formulation that separates cost, service, and emissions explicitly. A fifth is tighter integration with forecast-driven rolling-horizon planning and digital-twin infrastructure.

The expanded manuscript is therefore best interpreted as a full methodological foundation. It is already suitable for submission to outlets that accept mathematically grounded transportation or supply-chain methodology with controlled computational evidence. For journals demanding strong empirical validation, it should be treated as the methodological core of a future case-study extension.

11 Conclusion

This paper developed a fractional stochastic transportation model with Caputo memory, scenario-dependent demand, carbon trading, and CVaR-based risk aversion. The formulation generalizes conventional transportation and inventory-balance models by allowing present operational states to depend on a weighted history of past decisions and disturbances. In this sense, the model is designed for transportation systems in which history matters: congestion persists, recovery is delayed, deterioration accumulates, and service imbalance propagates across periods.

Several contributions were established. First, the paper integrated long-memory dynamics, stochastic recourse, and carbon-market decisions in a unified optimization framework. Second, the Caputo state equation was discretized through the L1 scheme to obtain a tractable deterministic equivalent. Third, basic theoretical properties were proved, including existence of an optimal solution, elimination of simultaneous buying and selling under a positive market spread, and stability with respect to moderate scenario perturbations. Fourth, a computational solution approach based on sample-average approximation and progressive hedging was presented. Fifth, an expanded benchmark analysis showed how the fractional order, demand volatility, carbon allowance, and risk aversion alter transportation performance in distinct ways.

The numerical evidence supports a clear interpretation. Stronger memory effects reshape shipment timing and protective behavior because the system internalizes a longer operational history. Greater volatility raises both expected cost and tail risk. Tighter allowance endowments create endogenous carbon purchases and stronger emissions pressure. Higher risk aversion improves shortage and CVaR performance, although with diminishing marginal gains. These findings confirm that fractional memory can be operationally meaningful in sustainable transportation planning.

The manuscript also clarifies where future work is needed. Real-data calibration of the fractional order is essential for empirical deployment. Robust or distributionally robust uncertainty models may be preferable when scenario probabilities are difficult to estimate. Endogenous carbon-price dynamics, multi-objective sustainability analysis, and digital-twin integration offer further extensions. In this sense, the present paper should be viewed not as the final word on fractional transportation modeling, but as a foundation for a broader research agenda at the interface of fractional dynamics, stochastic optimization, and sustainable logistics.

Overall, the paper argues for a simple but consequential idea: transportation planning should recognize memory. When operational history affects present performance, integer-order models

can be too forgetful. Caputo-based fractional dynamics provide a mathematically disciplined way to restore that memory and to connect it with uncertainty, risk, and environmental regulation in one coherent planning framework.

A Notation summary

For the reader's convenience, this appendix collects the principal notation used in the manuscript.

Table 6: Main notation used in the fractional stochastic transportation model.

Symbol	Meaning
I	Set of suppliers.
J	Set of customers or demand zones.
K	Set of transportation modes.
$T = \{1, \dots, N\}$	Set of planning periods.
Ω	Set of demand scenarios.
p_ω	Probability of scenario ω .
S_{it}	Supply available at supplier i in period t .
d_{jt}^ω	Demand of customer j in period t under scenario ω .
c_{ijkt}	Unit transportation cost on supplier–customer–mode arc (i, j, k) in period t .
e_{ijkt}	Unit emissions factor on arc (i, j, k) in period t .
Q_k	Capacity coefficient of transportation mode k .
f_k	Fixed activation cost of mode k .
h_j	Cost coefficient associated with the effective state at customer j .
π_j	Unit shortage penalty for customer j .
A	Initial allowance endowment in the carbon market.
p^b	Carbon-allowance buying price.
p^s	Carbon-allowance selling price.
ρ_j	Dissipation or deterioration parameter in the fractional state equation.
γ_j	Minimum required service fraction for customer j .
α	Fractional order in the Caputo derivative, with $0 < \alpha < 1$.
β	Confidence level used in the CVaR risk measure.
λ	Weight assigned to the CVaR term in the objective.
y_k	Binary variable indicating whether mode k is activated.
x_{ijkt}^ω	Shipment from supplier i to customer j by mode k in period t under scenario ω .
z_{jt}^ω	Effective inventory–service state of customer j in period t under scenario ω .
u_{jt}^ω	Shortage of customer j in period t under scenario ω .
b^ω	Carbon allowances purchased under scenario ω .
s^ω	Carbon allowances sold under scenario ω .

Symbol	Meaning
C^ω	Total scenario-dependent operational cost.
E^ω	Total scenario-dependent emissions.
η	Auxiliary CVaR threshold variable.
ξ_ω	Auxiliary excess variable for CVaR linearization.
a_r	L1 discretization coefficient, $a_r = (r + 1)^{1-\alpha} - r^{1-\alpha}$.
Δt	Length of one planning interval.

The notation is deliberately kept compact so that the model can be adapted to richer network structures without changing its conceptual core.

B Scenario-generation protocol

This appendix documents the logic used to generate the benchmark scenarios. The objective is not to replicate a specific industrial dataset but to define a reproducible synthetic environment consistent with the model assumptions.

B.1 Continuous-time demand representation

For each customer j , demand is generated from a drift–diffusion process of the form

$$dD_j(t) = \mu_j(t) dt + \sigma_j(t) dW_j(t), \quad (29)$$

where the drift component captures trend and the volatility term captures random fluctuations. In the benchmark design, the drift is allowed to vary moderately across customers to reflect heterogeneous market growth rates.

B.2 Discretization

On the planning grid $t_n = n\Delta t$, the process is discretized through an Euler-type scheme,

$$D_{j,n+1} = D_{j,n} + \mu_{j,n}\Delta t + \sigma_{j,n}\sqrt{\Delta t}\varepsilon_{j,n}, \quad (30)$$

where $\varepsilon_{j,n}$ are standard normal innovations. Cross-customer correlation can be incorporated through a Cholesky factorization of the innovation covariance matrix. In the benchmark, moderate positive dependence is introduced to emulate common market shocks.

B.3 Volatility scaling

To study sensitivity, the volatility coefficients are multiplied by a scalar factor $\nu \in \{0.50, 0.75, 1.00, 1.25, 1.50\}$. This creates nested families of demand trajectories with increasing uncertainty while preserving the same drift structure. The resulting scenarios are therefore comparable in a controlled way.

B.4 Training and validation samples

For each parameter regime, two independent samples are used:

- a training sample used to solve the SAA problem;
- an independent validation sample used to evaluate the recovered policy out of sample.

This split is important because a finite scenario sample can lead to optimistic in-sample results. Validation therefore functions as a guard against overfitting.

B.5 Interpretation of synthetic scenarios

The benchmark scenarios should be interpreted as controlled stress patterns rather than forecasts of a single real system. Their purpose is to reveal how the optimization model reacts when uncertainty, memory, and carbon constraints interact. In later empirical work, the same protocol can be replaced by historical residual resampling, parametric bootstrap, or forecast-distribution sampling from a predictive model.

B.6 Why a synthetic benchmark is still useful

Methodological papers often face a trade-off between realism and transparency. A proprietary case study may be realistic but difficult to reproduce. A synthetic benchmark may be less realistic but easier to audit and extend. The present paper adopts the second path deliberately. By making the scenario logic explicit, the manuscript allows later researchers to replace only the data layer while preserving the full fractional stochastic optimization architecture.

C Additional proof details and analytical remarks

This appendix supplements the proofs in the main text with additional remarks that may be useful for readers interested in the analytical structure of the model.

C.1 Boundedness of the effective state

In the existence proof, boundedness of the effective state variable z_{jt}^ω follows from the finite-horizon structure together with bounded shipments, finite demand scenarios, and nonnegative shortage recourse. To see this more concretely, note that the discretized fractional balance at period n can be rearranged as

$$(\kappa_{n,n-1} + \rho_j) z_{jn}^\omega = \sum_{i,k} x_{ijkn}^\omega - d_{jn}^\omega + u_{jn}^\omega + \sum_{r=0}^{n-2} \kappa_{n,r} (z_{j,r+1}^\omega - z_{jr}^\omega) + \kappa_{n,n-1} z_{j,n-1}^\omega. \quad (31)$$

Because all coefficients are finite and positive and because the horizon is finite, recursive bounds can be constructed period by period. This gives a direct state-boundedness argument that complements the compactness proof.

C.2 Interpretation of complete recourse

The scenario-stability theorem relies on complete recourse. In the present context, this means that any feasible first-stage decision can be supplemented by second-stage shortage variables that absorb unmet demand. This is operationally reasonable because shortage is explicitly allowed and penalized. Complete recourse therefore does not claim that service can always be maintained perfectly; it claims that the model remains feasible because unmet demand can be recorded and penalized rather than making the optimization problem infeasible.

C.3 Memory and perturbation propagation

A useful intuition behind the stability theorem is that the fractional operator spreads perturbation over time but does not amplify it without bound on a finite horizon. The L1 coefficients are summable and positive. Therefore a bounded change in one period's demand produces a bounded cumulative change in later effective states. This is the essential mechanism behind the Lipschitz-type value bound.

C.4 Potential extension: convexity under fixed first-stage decisions

If the first-stage activation vector is fixed, the remaining model becomes a continuous optimization problem with linear constraints and a linearized CVaR term. Without PH penalties it is therefore a linear program in the discretized setting. With PH penalties it becomes a convex quadratic program. This observation explains why scenario decomposition is attractive: the difficulty of the problem lies in the first-stage integrality and in the multiplication of scenario blocks, not in nonconvex recourse once the first stage is fixed.

C.5 Remark on variable-order memory

If the fractional order becomes time dependent or customer specific, the positivity of the discretization weights can still hold under standard constructions, but the clean monotonic decay property may be altered. This would complicate both analysis and interpretation. The present manuscript therefore focuses on constant-order Caputo memory so that the meaning of persistence remains transparent.

C.6 Analytical takeaway

The broader analytical lesson is that introducing fractional memory changes the problem's structure primarily through temporal density, not through pathological infeasibility or loss of finite-dimensional control. This makes the model mathematically richer than a classical transportation program, but still tractable enough for rigorous optimization analysis.

D Calibration and implementation roadmap

This appendix outlines how the proposed model could be calibrated and deployed in a real transportation setting. The goal is to make the manuscript more directly useful for empirical follow-up work.

D.1 Data requirements

A practical implementation would ideally assemble four data blocks:

- D1.** shipment and order history by customer, period, and mode;
- D2.** service outcomes such as late deliveries, backlog, or stockout observations;
- D3.** emissions and carbon-accounting factors by mode or vehicle class;
- D4.** capacity and procurement information for transport modes and suppliers.

These blocks are sufficient to calibrate both operational costs and the stochastic-demand layer. The most delicate calibration task is the fractional order.

D.2 Estimating the fractional order

There are several possible approaches to estimating α .

- **Autocorrelation decay matching:** estimate the empirical decay pattern of inventory discrepancy, congestion index, or service imbalance, then choose the fractional order whose implied memory kernel best matches the observed decay.
- **Prediction-error minimization:** compare one-step-ahead or multi-step-ahead prediction performance of integer-order and fractional-order state equations on historical data.
- **Indirect structural calibration:** estimate α jointly with dissipation parameters by minimizing the gap between observed and simulated system states over time.

No single method is universally best. The appropriate choice depends on data quality and whether the effective state variable is directly observed or latent.

D.3 Calibration of carbon-market parameters

Allowance endowment, buying price, and selling price can usually be obtained from regulatory or market data. The more difficult part is estimating unit emissions factors e_{ijkt} . These should reflect the operational granularity of the model. If the model distinguishes modes only coarsely, then emissions factors should also be aggregate. If the model distinguishes vehicle classes, load-dependent emissions, or route distances, then the emissions layer should be refined accordingly.

D.4 Deployment protocol

A reasonable deployment pathway consists of five stages:

- P1.** build a historical backtest environment preserving decision chronology;
- P2.** calibrate the demand process and candidate fractional orders;
- P3.** solve the model on historical rolling windows or scenario sets;

- P4.** compare outcomes against the incumbent planning policy;
- P5.** validate service, cost, and emissions performance with managers.

This protocol turns the present methodological paper into a directly testable empirical research agenda.

D.5 Journal-readiness implication

Including a calibration roadmap matters for submission. Reviewers often ask whether a methodological model can be used in practice. By specifying data requirements, estimation logic, and deployment steps, the paper demonstrates that the proposed fractional stochastic model is not purely theoretical. It is ready to be transferred into a case-study setting when the relevant operational data become available.

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